Deep-Unfolded Compressed Sensing for Energy-Efficient Wearable ECG: Diagnostic Preservation, Learned Binary Sensing, and Cross-Dataset Generalization

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***Abstract*—Compressed sensing (CS) lowers the radio energy of wearable electrocardiogram (ECG) sensors by transmitting far fewer samples, but the value of the system hinges on how faithfully–and how cheaply–the signal is reconstructed at the receiver. We present a deployment-oriented empirical study of a compact 1-D deep-unfolded reconstruction network (an ISTA-Net^{+}-style unrolling) for wearable ECG, evaluated under a strict inter-patient protocol. Against a carefully tuned classical solver (FISTA with a wavelet basis) and a $25\times$ larger black-box convolutional network, the unrolled network with only $6{,}451$ parameters attains 8.0% PRDN at a $4{:}1$ compression ratio (vs. 28.1% and 35.9%), and its advantage widens at aggressive compression (24.3% vs. 81.9% at $10{:}1$). Against *published* deep CS-ECG decoders it is 38–$92\times$ smaller while matching or bettering their reported distortion; protocols differ, so we do not claim state-of-the-art, but most of the quality is clearly reachable at a fraction of the model size. Learning the sensing matrix nearly halves the distortion ($8.0 \rightarrow 4.3\%$), and constraining it to hardware-realizable *binary* (± 1) entries costs little (4.6%) while replacing every measurement multiplier with an adder; a fixed-random ± 1 control arm (8.3%) shows the gain comes from learning, not from binarizing. Crucially, the *diagnosis* survives compression: an AAMI beat classifier loses only 1.2 points of accuracy on unrolled reconstructions versus 10.6 for the classical solver, and V-beat precision degrades least. The learned prior transfers zero-shot to PTB-XL for 0.5 PRDN points once a resampling-matched control is used. A transmit-energy model shows a 75% radio-energy reduction at $r{=}0.25$. The study is a systems contribution rather than a new operator, and its artifacts are reproducible on one commodity GPU.**

***Keywords*—compressed sensing, deep unfolding, electrocardiogram, wearable sensors, energy efficiency, algorithm unrolling, inter-patient.**

I. Introduction

Continuous ECG monitoring on wrist- and patch-based wearables is limited less by sensing than by *communication*: streaming the raw waveform over a low-power radio dominates the energy budget of a body-area sensor node. Compressed sensing (CS) [1], [2] offers a principled remedy–acquire a small number of random linear measurements $\mathbf{y} = \mathbf{\Phi x}$ and recover $\mathbf{x}$ by exploiting sparsity–so the node transmits $m \ll N$ values per window and the reconstruction burden moves to the receiver. For wearable ECG this directly trades a few percent of reconstruction fidelity for a large reduction in transmit energy [3].

The reconstruction algorithm therefore decides whether CS is worthwhile. Classical convex solvers (e.g., FISTA [4] over a wavelet basis) are interpretable but need hundreds of iterations and degrade sharply as the measurement ratio drops. Generic "black-box" neural networks that regress the signal from measurements are fast but data-hungry and prone to inter-patient overfitting. *Deep unfolding* (algorithm unrolling) [5], [6] sits between the two: it turns a fixed number of solver iterations into network layers, retaining the physics of the forward model while learning the proximal operator. Unrolled CS is mature for images [7]–[9] and has already been applied to ECG [10]–[12]; the method itself is not new.

Contribution. This paper does *not* propose a new reconstruction operator. It contributes a rigorous, deployment-oriented empirical study of a compact 1-D unrolled network for wearable ECG that, under a strict inter-patient protocol, (i) quantifies the rate–distortion, diagnostic, data-efficiency and efficiency trade-offs against a *tuned* classical solver and a larger black-box network; (ii) shows that constraining the learned sensing matrix to a *hardware-realizable* binary (± 1) form–implementable with adders alone, without analog multipliers–retains nearly all the benefit of a dense learned one; (iii) demonstrates that the clinical task (arrhythmia beat classification) survives aggressive compression, using a downstream classifier rather than only waveform error; (iv) measures zero-shot cross-dataset generalization to PTB-XL against a resampling-matched control; and (v) reports

a concrete transmit-energy model. Prior work has already unrolled CS solvers for ECG and even learned the sensing operator [12]; the novelty claimed here is the hardware-constrained sensing and the deployment-oriented evaluation, not the architecture. Code, seeds and result tables will be released.

II. Related Work

A. Algorithm unrolling

Since LISTA [5], unrolling iterative solvers into trainable networks has become a standard bridge between model-based and learned inversion [6]. ISTA-Net and ISTA-Net$^+$ [7] learn a nonlinear sparsifying transform with a symmetry constraint; ADMM-CSNet [8] and AMP-Net [9] unroll ADMM and approximate message passing. These target natural images; we adopt the ISTA-Net$^+$ template in 1-D for physiological signals and study it as a system.

B. CS for ECG

CS was proposed for energy-efficient wireless ECG by Mamaghanian *et al.* [3], and the field has since moved from fixed sparsifying bases toward learned reconstruction. CSNet [10] regresses the signal with a CNN–LSTM decoder and Hua *et al.* [11] use an inception/LSTM decoder, both with $\sim 6\times10^5$ parameters; Unni *et al.* [13] instead place a learned denoiser inside a classical solver. Closest to this work, Erkoc [12] unrolls an ISTA-style solver with a learnable step size, *learns the sensing operator*, and uses dilated-CNN proximal blocks with adaptive gating (2.45×10^5 parameters), evaluated on a subject-disjoint MIT-BIH split using PRD and SNR.

C. Position of this paper

We therefore claim neither the unrolled architecture, nor the learnable step size, nor a learned $\mathbf{\Phi}$ as new–all of these already appear in [12]. This study is deliberately *orthogonal*: it asks how far an *extremely compact* unrolled decoder can be pushed, whether a learned sensing matrix can be made *hardware-realizable*, and whether the *diagnosis*–not just the waveform–survives compression. Distortion is reported with the PRD, whose clinical categories were established by Zigel *et al.* [14]; inter-patient evaluation and the AAMI classes follow de Chazal [15] on MIT-BIH [16], and cross-dataset tests use PTB-XL [17]. Bipolar ±1 sensing is trained with a straight-through estimator [18].

III. Method

A. CS forward model

Each ECG window $\mathbf{x}\in\mathbb{R}^N$ (N=512, $\sim$1.42 s at 360 Hz) is sensed as $\mathbf{y}=\mathbf{\Phi}\mathbf{x}$ with $\mathbf{\Phi}\in\mathbb{R}^{m\times N}$ and measurement ratio $r{=}m/N$; the compression ratio is $\mathrm{CR}{=}1/r$. The decoder is initialized with the transpose $\mathbf{x}_0=\mathbf{\Phi}^\top\mathbf{y}$.

B. Unrolled reconstruction network

Fig. 1 shows the complete sensing-to-reconstruction pipeline and the internal structure of one unrolled phase. We unroll K ISTA iterations. Phase k applies a data-consistency gradient step followed by a learned proximal step:

$$\mathbf{r}_k=\mathbf{x}_{k-1}-\rho_k\,\mathbf{\Phi}^\top(\mathbf{\Phi}\mathbf{x}_{k-1}-\mathbf{y}), \tag{1}$$

$$\mathbf{x}_k=\mathbf{r}_k+G_k\big(\mathrm{soft}(F_k(\mathbf{r}_k),\theta_k)\big), \tag{2}$$

with $\mathrm{soft}(u,\tau)=\mathrm{sign}(u)\,\mathrm{ReLU}(|u|-\tau)$. F_k, G_k are small 1-D convolutional blocks (two layers, 32 channels, kernel 3), shared across phases; ρ_k,θ_k are learned per phase. The step size is initialized at the stable value $\rho_k{=}1/L$, where L is the spectral norm of $\mathbf{\Phi}^\top\mathbf{\Phi}$, which we found essential for stability. Following ISTA-Net$^+$, a symmetry term $\sum_k\|G_k(F_k(\mathbf{r}_k))-\mathbf{r}_k\|^2$ encourages F, G to be inverse. The loss is $\mathcal{L}=\|\hat{\mathbf{x}}-\mathbf{x}\|^2+\gamma\,\mathcal{L}_{\text{sym}}$ with $\gamma{=}0.01$. With $K{=}9$ shared phases the network has only 6,451 parameters.

C. Learned and hardware-constrained sensing

We compare three choices of the sensing matrix $\mathbf{\Phi}$.

Fixed. A random Gaussian matrix with ℓ_2-normalized rows. This is the standard CS choice and is used in all experiments unless stated otherwise.

Learned (dense). $\mathbf{\Phi}$ becomes a trainable layer and is optimized jointly with the decoder, so the measurements adapt to ECG statistics instead of being random.

Learned (binary). A dense $\mathbf{\Phi}$ needs one analog multiplier per entry, which is costly on a low-power node. We therefore restrict the entries to ±1, which turns every measurement into a signed accumulation. Since $\mathrm{sign}(\cdot)$ has zero gradient almost everywhere, we train through it with a straight-through estimator [18]: a real-valued latent $\mathbf{\Phi}_\ell$ is stored, the forward pass uses the binarized matrix

$$\bar{\mathbf{\Phi}}=\tfrac{1}{\sqrt{N}}\,\mathrm{sign}(\mathbf{\Phi}_\ell), \tag{3}$$

whose scaling keeps the rows unit-norm, and the backward pass passes gradients through unchanged, i.e. $\partial\bar{\mathbf{\Phi}}/\partial\mathbf{\Phi}_\ell\approx\mathbf{I}$. The resulting front-end needs only sign flips and adders: the $m\cdot N$ multiply–accumulates per window become $m\cdot N$ additions.

IV. Experimental Setup

A. Data and protocol

We use the MIT-BIH Arrhythmia Database [16] (lead MLII, 360 Hz) with the de Chazal inter-patient split [15]: DS1 (22 records) for training, DS2 (22 records) for testing, disjoint by patient (asserted); DS2 provides 27,918 non-overlapping test windows. Windows are z-scored per window. Cross-dataset tests use PTB-XL [17] (lead II, resampled $100\to360$ Hz), zero-shot.

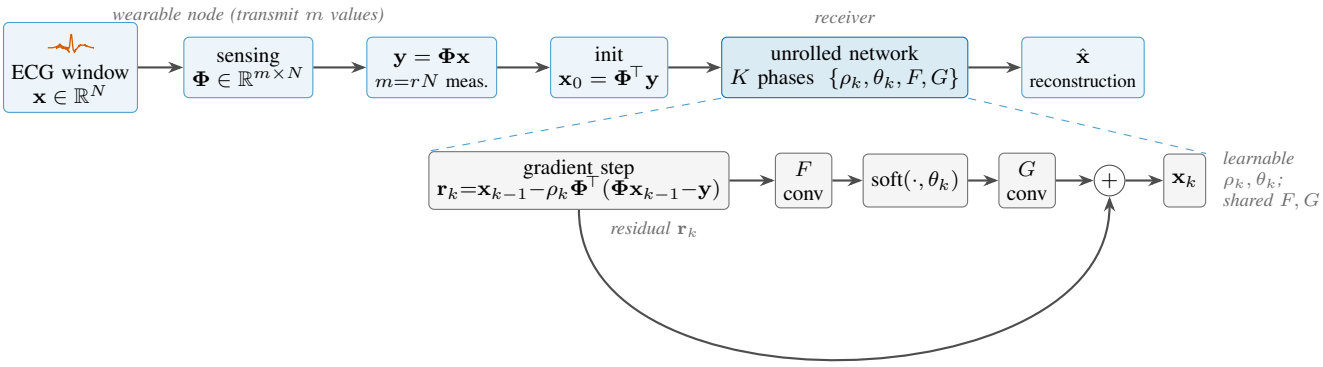


Fig. 1. CS wearable-ECG pipeline and unrolled reconstruction network. The node transmits only $m{=}rN$ measurements $\mathbf{y} = \mathbf{\Phi x}$; the receiver runs K phases, each a gradient step plus a learned proximal operator (F/G with a soft threshold). Step sizes ρ_k and thresholds θ_k are learned per phase; $\mathbf{\Phi}$ is fixed-random, learned, or hardware-realizable binary.

B. Metrics and PRD conventions

Two PRD conventions coexist in the CS-ECG literature and are not interchangeable. For a window $\mathbf{x}$ with mean $\bar{x}$ we use

$$\mathrm{PRDN} = 100\,\frac{\|\mathbf{x}-\hat{\mathbf{x}}\|}{\|\mathbf{x}-\bar{x}\|}, \qquad \mathrm{PRD}_{\mathrm{raw}} = 100\,\frac{\|\mathbf{x}-\hat{\mathbf{x}}\|}{\|\mathbf{x}\|}. \tag{4}$$

Zigel's clinical categories [14] are defined on the mean-removed PRDN, whereas the decoders we compare against [10]–[12] report raw PRD. *Every value here is computed directly from the reconstructions in the convention named in its heading; none is converted.* The two differ by the per-window factor $\sigma/\sqrt{\sigma^2+\mu^2}$ (σ, μ the window std and mean), whose median over the 27,918 DS2 windows is 0.81 (IQR 0.58–0.93), so the convention must be stated wherever a value is compared. We use PRDN by default and raw PRD only in Table IV. We further report SNR, R-peak F1, and–as the diagnostic endpoint–*absolute* accuracy and macro-F1 of an AAMI (N/S/V/F/Q) beat classifier together with the drop relative to originals. Efficiency is measured by parameters, FLOPs and CPU/GPU latency; energy by a transmit model.

C. Downstream beat classifier

Because the diagnostic endpoint depends on it, the classifier is specified in full. *Segmentation:* each beat is a 256-sample (0.71 s) window centred on the R-peak. R-peak locations are taken from the MIT-BIH *reference annotations* and are identical for the original and for every reconstruction, so the comparison isolates beat *morphology* and is not confounded by detector failures on the reconstructed signal; detection quality is reported separately as QRS-F1. *Features:* none – the raw beat window, z-scored per beat. *Architecture:* three 1-D convolutions (16/32/64 channels, kernels 7/5/3, ReLU), max-pooling after the first two, global max-pooling, and a linear layer over the five AAMI classes. *Training:* original DS1 beats only, Adam, lr 10^{-3}, batch 256, 20 epochs, cross-entropy weighted inversely to class frequency. The trained classifier is then applied unchanged to original and reconstructed DS2 beats.

D. Baselines

Classical: FISTA minimizing $\frac{1}{2}\|\mathbf{y}-\mathbf{\Phi x}\|^2 + \lambda\|\Psi\mathbf{x}\|_1$ with an orthonormal DCT or Daubechies-4 basis; λ is grid-searched and the better basis (db4) selected. *Black-box:* a 1-D U-Net (162,657 parameters) mapping $\mathbf{\Phi}^\top\mathbf{y} \to \mathbf{x}$, sized *larger* than the unrolled net so the parameter-efficiency comparison is not stacked. A sweep over weight decay (0 to $5{\times}10^{-3}$) and dropout (0 to 0.3) moved its test PRDN by only 0.33 points (35.65–35.98%), so its weak inter-patient generalization is *architectural*–it does not exploit the sensing physics–rather than under-tuning.

E. Implementation

PyTorch on a single consumer GPU (16 GB). Adam, cosine schedule, batch 128; three random seeds for learned models (mean±std). All methods share the same $\mathbf{\Phi}$, except where $\mathbf{\Phi}$ is itself the variable under study (Section V-B).

V. Results

A. Rate–distortion and diagnostic preservation

Fig. 2 plots distortion and diagnostic quality against the measurement ratio, and Table I lists the operating point at $r{=}0.25$. On the rate–distortion axis of Fig. 2(a) the unrolled network dominates at every ratio, and *its margin is largest at aggressive compression*: at $r{=}0.1$ it holds 24.3% PRDN while FISTA collapses to 81.9%; at $r{=}0.25$ it reaches 8.0% vs. 28.1%. Classical CS becomes competitive only when measurements are plentiful ($r \geq 0.5$)–precisely the regime with little energy benefit. The black-box network, by contrast, barely improves with more measurements (51% → 26% across a 7× range), evidence that it ignores the sensing physics and overfits across patients.

The same ordering holds for the diagnostic metric in Fig. 2(b): at $r{=}0.1$ the unrolled net keeps a QRS-F1 of

TABLE I
MAIN COMPARISON AT r=0.25

Method	PRDN (%)	SNR (dB)	QRS-F1	Params[a]	CPU[b] (ms)
FISTA+db4	28.1	11.0	0.963	–	41.3
Black-box U-Net	35.9	8.9	0.935	162,657	1.0
Unrolled (ours)	**8.0**	**21.9**	**0.982**	**6,451**	2.9

[a]Trainable parameters. [b]Per-window CPU latency.

TABLE II
ABSOLUTE AND PER-CLASS BEAT CLASSIFICATION AT r=0.25

Beats from	Acc. (%)	M-F1 (%)	ΔAcc (pt)	ΔF1 (pt)	Class S		Class V	
					Se	PPV	Se	PPV
Original	74.8	33.1	–	–	9.6	3.2	83.0	63.3
Unrolled (ours)	**73.6**	**32.2**	**1.2**	**0.9**	7.3	2.4	**83.2**	**59.8**
FISTA+db4	64.2	28.5	10.6	4.5	17.9	4.2	84.5	43.2
Black-box U-Net	70.8	27.0	4.1	6.0	3.5	1.8	77.1	34.3

Inter-patient DS2, 49,693 beats; classes N, F, Q omitted.
Se: sensitivity, PPV: predictive value; R-peaks from annotations.

0.955 while FISTA falls to 0.716. A Wilcoxon signed-rank test on per-window PRD (n=6000, r=0.25) confirms the unrolled net is significantly better than both baselines ($p \approx 0$). Inspecting the reconstructions explains why: on a window at the median PRDN the unrolled trace overlays the original through the QRS complex, whereas FISTA rings along the baseline and the black-box oversmooths the morphology.

B. Learned and binary sensing

Fig. 3 compares the four sensing matrices obtained by crossing {fixed, learned} with {dense Gaussian, binary ± 1}. The full 2×2 design is what separates the two things a learned ± 1 front-end does at once: *binarizing* $\mathbf{\Phi}$, which is what makes it hardware-realizable, and *learning* $\mathbf{\Phi}$, which is what adapts it to ECG statistics.

The two effects separate cleanly. *Binarizing* $\mathbf{\Phi}$ is nearly free: a fixed random ± 1 matrix costs 0.31 PRDN points against a fixed Gaussian one (8.33 vs. 8.02%), and once $\mathbf{\Phi}$ is learned the ± 1 constraint is not separable from seed noise (4.55 ± 0.62 vs. $4.29 \pm 0.13\%$). *Learning* $\mathbf{\Phi}$ is the large lever, removing 3.73 points when dense ($8.02 \rightarrow 4.29\%$) and 3.78 when binary ($8.33 \rightarrow 4.55\%$), far beyond that spread. The two are therefore almost independent and additive, which settles the question directly: the hardware-realizable front-end *does* need learning–a fixed ± 1 matrix sits at 8.33%, beside the fixed Gaussian and far from the 4.55% of the learned ± 1. At r=0.1 the fixed arms behave the same way (binarizing costs 0.84 points, $24.39 \rightarrow 25.23\%$) and learning again dominates, but there the ± 1 constraint actively *helps*: the learned ± 1 matrix reaches $9.16 \pm 0.02\%$ against $12.14 \pm 0.89\%$ for learned dense. The binary constraint appears to stabilize the joint optimization at aggressive compression–the dense arm swings by almost a point across seeds while the binary arm is essentially seed-invariant. A learned ± 1 sensing matrix therefore retains all of the benefit of a learned dense one, and more at low r, while replacing every measurement multiplier with an adder.

C. Data efficiency

The physics prior also substitutes for labels. Trained on only 5% of the DS1 windows (1,256 of 25,126) the unrolled net already reaches 23.7% PRDN–better than the black-box trained on *all* the data (35.9%)–and improves monotonically to 19.6, 13.6, 10.1 and 8.0% at 10, 25, 50 and 100%, whereas the black-box moves only from 39.5 to 35.9% over the same $20\times$ range.

D. Does the diagnosis survive? (beat classification)

Waveform error can mislead; we therefore test whether *arrhythmia classification* survives compression. The classifier of Section IV-C, trained on original DS1 beats, is applied to DS2 beats extracted from the originals and from each reconstruction. Table II reports absolute accuracy and macro-F1, the drop relative to the originals, and per-class sensitivity (Se) and positive predictive value (PPV) for the two clinically important minority classes, S and V.

The unrolled network costs only a 1.2-point accuracy drop (and 0.9 macro-F1), versus 10.6 points for FISTA–whose high-frequency ringing corrupts beat morphology–and 4.1 for the black-box; diagnosis is essentially preserved through 4:1 compression.

The absolute numbers add context the drop alone hides, and we report them plainly. Macro-F1 is low *even on the original signal* (33.1%): S beats are rare (1,837 of 49,693 DS2 beats) and morphologically heterogeneous across patients, and this classifier detects them barely above chance *before any compression* (Se 9.6%, PPV 3.2%). The S column therefore characterizes the classifier's limits rather than the reconstruction's, and no conclusion should be drawn from it. The informative per-class endpoint is V, which the classifier does learn: V-beat PPV falls from 63.3% on originals to 59.8% on unrolled reconstructions, but to 43.2% for FISTA and 34.3% for the black-box. The ordering of the waveform metrics is thus reproduced on a clinically meaningful per-class endpoint, which is the claim we make; the *drop*, holding the classifier fixed, remains the statistic that isolates what compression destroys.

E. Cross-dataset generalization

A learned reconstruction prior is only useful if it does not overfit the training database. We therefore apply the MIT-BIH-trained models *zero-shot* to 150 PTB-XL records (lead II), never seen in training; the classical solver is dataset-agnostic and is allowed to re-tune λ on a PTB-XL split for fairness.

PTB-XL is recorded at 100 Hz and must be resampled to the 360 Hz the models expect, which band-limits it to 50 Hz and leaves it measurably smoother–and hence easier to reconstruct–than native MIT-BIH. A raw PTB-XL-vs-MIT-BIH comparison would therefore confound dataset transfer with resampling. We control for this by passing MIT-BIH DS2 through the *identical* $360 \rightarrow 100 \rightarrow 360$ Hz chain and re-

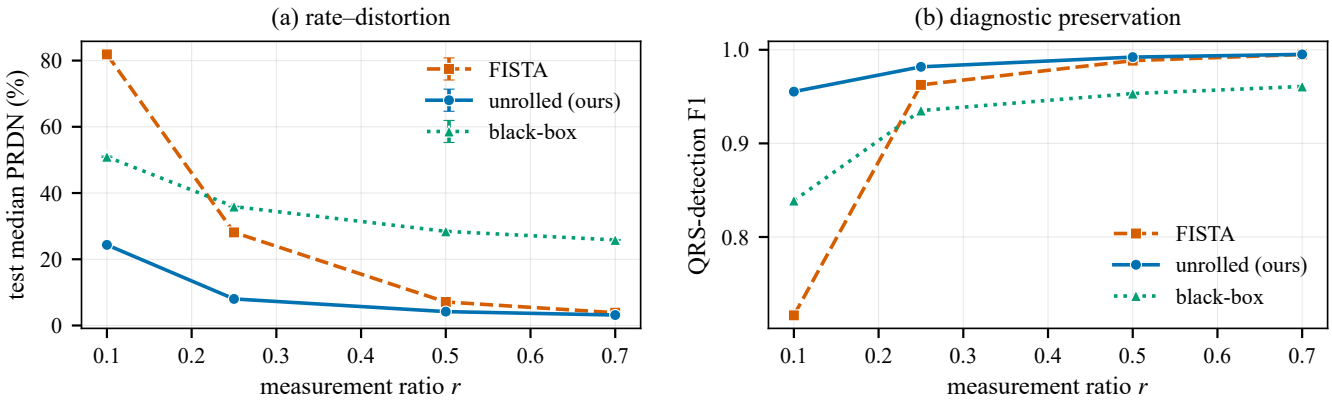


Fig. 2. Inter-patient (DS2) rate–distortion and diagnostic preservation. (a) median PRDN vs. measurement ratio r (error bars: 3 seeds); (b) QRS-detection F1. The unrolled network (ours) leads at all ratios and most decisively under aggressive compression, where classical CS collapses.

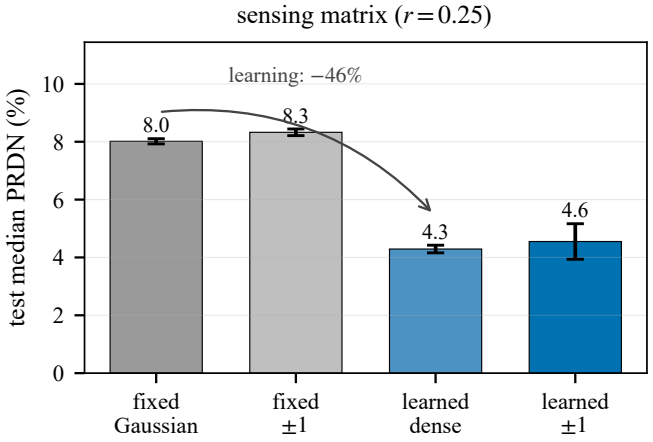


Fig. 3. The four sensing matrices at r=0.25 (3 seeds). Crossing {fixed, learned} with {dense, ±1} separates the effect of binarizing $\mathbf{\Phi}$ from the effect of learning it.

TABLE III
ZERO-SHOT CROSS-DATASET GENERALIZATION

Method	DS2 native	DS2 b.-lim.[a]	PTB-XL 0-shot
FISTA+db4[b]	27.9	26.9	31.5
Black-box U-Net	36.1	35.8	37.3
Unrolled (ours)	**8.6**	**7.5**	**8.0**

[a]DS2 through the same 360→100→360 Hz chain PTB-XL needs.
[b]λ re-tuned on a PTB-XL split. Median PRDN (%), r=0.25.
All columns use one checkpoint, so the comparison is paired and its native DS2 value differs slightly from the 3-seed mean in Table I.

TABLE IV
PUBLISHED DEEP CS-ECG DECODERS AT SR=0.10

Method	Params[a]	PRD[b] (%)	learned $\mathbf{\Phi}$	±1 $\mathbf{\Phi}$	diagn. endpt.
CSNet [10]	595,937	14.2	no	no	no
Hua *et al.* [11]	596,897	11.5	yes	no	no
Erkoc [12]	245,069	7.7	yes	no	no
Ours (learned dense)	6,451	8.9	yes	no	yes
Ours (learned ±1)	**6,451**	**6.9**	yes	**yes**	**yes**

[a]At SR=0.25; [10], [11] as re-implemented in [12].
[b]Raw (non-mean-removed) PRD, the convention used by these works. Protocols differ: indicative, not controlled.

evaluating the same models, which isolates the band-limiting effect; the middle column of Table III is that control.

The confound is real but small. Band-limiting alone makes DS2 easier by 1.1 PRDN points for the unrolled net (8.6→7.5%), so the correct comparison is PTB-XL (8.0%) against *band-limited* DS2 (7.5%). Read that way, zero-shot transfer to a different database, patient population and acquisition hardware costs 0.5 points–small, but *not* the exact parity the naive 7.99 vs. 8.0% comparison suggested, which was partly a resampling artifact. The ranking is unaffected: the unrolled net still beats the re-tuned FISTA and the black-box by a wide margin on PTB-XL.

F. Comparison with published CS-ECG decoders

The baselines above are ones we control; Table IV places our decoder against *published* deep CS-ECG models at their *sensing rate* (SR, identical to r). These works report raw PRD, so this is the one table in that convention, and our entry comes from evaluating our own reconstructions under the raw definition in (4)–*not* from rescaling a PRDN value.

With a learned dense $\mathbf{\Phi}$ our 6,451-parameter decoder reaches 8.9% at SR=0.10–better than CSNet (14.2%) and Hua *et al.* (11.5%) with 92× fewer parameters, and behind the dilated-CNN model of Erkoc [12] (7.7%, 38× more parameters). With the hardware-constrained ±1 operator it reaches 6.9%, below all three. We nevertheless *do not claim state-of-the-art distortion*: window length, record split and PRD definition differ across these works, so Table IV is indicative rather than controlled. The robust point is that most of the quality is reachable at a fraction of the model size, from a decoder that also accepts a ±1 sensing matrix and preserves the diagnosis.

G. Efficiency and energy

Reconstruction cost. The unrolled network is 14× faster than FISTA on CPU (2.9 vs. 41.3 ms per window) *and* 3.5× more accurate, so it dominates the classical solver on both

TABLE V
ABLATIONS AND TRANSMIT-ENERGY MODEL

Ablation (r=0.25)			**Transmit energy**		
config	**params**	**PRDN (%)**	r	**mJ/h**	**save**
K=5	6,443	11.7	1.00	77.8	–
K=9	6,451	8.0	0.50	38.9	50%
K=13	6,459	7.3	0.25	19.4	75%
per-phase	57,915	8.1	0.10	7.8	90%

axes. The black-box network is the fastest–a single feed-forward pass–but its quality is unusable at this ratio, and it needs $25\times$ more parameters (162.7k vs. 6.5k).

Transmit energy. On the sensor side, the per-window radio energy is $E_{\text{tx}} = m\, b\, e_{\text{bit}}$, where b=12 bits per sample and e_{bit} is the radio energy per bit. At f_s=360 Hz a node emits $f_s \cdot 3600/N \approx 2531$ windows per hour. Because only $m = rN$ values are transmitted, the saving is $1-r$ regardless of the constants: at r=0.25 the node uses **75% less** transmit energy (19.4 vs. 77.8 mJ/hour), and at r=0.10, 90% less (Table V, right).

H. Ablations

Table V (left) shows more unrolling phases help with negligible parameter cost (weights shared), saturating near K=9–13; *per-phase* weights use $9\times$ more parameters for no gain, validating weight sharing. The symmetry term γ was near-neutral on ECG (7.96% vs. 7.98%).

VI. DISCUSSION AND LIMITATIONS

The results consistently favor the unrolled network, but the contribution is empirical rather than methodological. Distortion is reported in two conventions that must not be mixed (Section IV-B); each value is computed directly in the convention named in its heading. Our compact decoder records a lower raw PRD at SR=0.10 than the much larger published models of Table IV, but their protocols differ from ours, so we read that as indicative and claim parameter efficiency, hardware-realizable sensing and diagnostic preservation rather than minimum PRD. The beat classifier is a simple baseline–S-class detection is near chance even on originals–so the *drop*, which holds it fixed, is the statistic we rely on; a stronger classifier could change the absolute numbers. PTB-XL lead II remains a limb-lead proxy for MLII, and the energy figure is a model with stated constants, not an on-device measurement. Crucially, all results come from two open, pre-recorded databases; we do *not* evaluate a real-time acquisition pipeline or a physical low-power sensing front-end, so the binary-Φ and transmit-energy claims are established in simulation rather than measured on fabricated hardware. Ablations use a single seed. These bound the strength of the conclusions without changing them.

VII. CONCLUSION

For energy-constrained wearable ECG, a 6.5k-parameter deep-unfolded reconstruction network reaches a markedly better rate–distortion and, more importantly, *diagnostic* operating point than both a tuned classical solver and a much larger black-box network under a strict inter-patient protocol. Learning the sensing matrix nearly halves distortion and survives a hardware-realizable ± 1 constraint, whereas a *fixed* ± 1 matrix does not: the gain is in the learning, and binarization is what makes it deployable. The practical value of CS-ECG therefore lies not in a new operator but in the co-design of compact unrolled reconstruction, learned low-cost sensing, and *diagnosis-preserving* evaluation.

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